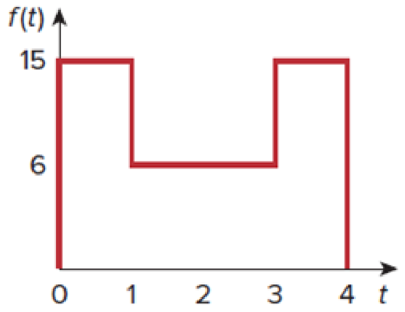
# Chapter 15, Problem 16P

## Problem

Obtain the Laplace transform of *f* (*t*) in Fig. 15.28.

**Figure 15.28:**




## Step-by-step solution

**Step 1** of 3

Refer to Figure 15.28 in the textbook.

The function $f(t)$ is defined as,

$$f(t)=\begin{cases}15 & 0<t<1\\ 6 & 1<t<3\\ 15 & 3<t<4\end{cases}$$

Determine the Laplace transform of $f(t)$.

Consider the general equation for Laplace transform.

$$F(s)=\int_0^{\infty} f(t)e^{-st}dt \quad \ldots\ldots (1)$$

**Step 2** of 3

Determine the Laplace transform of $f(t)$.

$$\begin{aligned}F(s)&=\int_0^4 f(t)e^{-st}dt\\ &=\int_0^1 f(t)e^{-st}dt+\int_1^3 f(t)e^{-st}dt+\int_3^4 f(t)e^{-st}dt\\ &=\int_0^1 15e^{-st}dt+\int_1^3 6e^{-st}dt+\int_3^4 15e^{-st}dt\\ &=15\int_0^1 e^{-st}dt+6\int_1^3 e^{-st}dt+15\int_3^4 e^{-st}dt\end{aligned}$$

**Step 3** of 3

Simplify the equation further,

$$
\begin{aligned}
F(s) &= 15\left|\frac{e^{-st}}{-s}\right|_0^1 + 6\left|\frac{e^{-st}}{-s}\right|_1^3 + 15\left|\frac{e^{-st}}{-s}\right|_3^4 \\
&= \frac{-1}{s}\left[15\left(e^{-s}-e^{0}\right)+6\left(e^{-3s}-e^{-s}\right)+15\left(e^{-4s}-e^{-3s}\right)\right] \\
&= \frac{-1}{s}\left[15e^{-s}-15+6e^{-3s}-6e^{-s}+15e^{-4s}-15e^{-3s}\right] \\
&= \frac{-1}{s}\left[9e^{-s}-9e^{-3s}+15e^{-4s}-15\right]
\end{aligned}
$$

Therefore, the Laplace transform of $f(t)$ is $\boxed{\frac{-1}{s}\left[9e^{-s}-9e^{-3s}+15e^{-4s}-15\right]}$.